2.3 Machine Learning vs. Deep Learning

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Machine Learning vs. Deep Learning: Picking the Right Tool for the Problem

Prior to this assignment, I often used the terms "deep learning" and "machine learning" interchangeably. Although they act in somewhat different ways, they are connected since deep learning is a part of machine learning. In standard machine learning, an algorithm such as a decision tree or support vector machine is fed the features that are considered important. Deep learning utilizes multi-layered neural networks that independently identify key characteristics in raw data. I choose image identification using a convolutional neural network (CNN) as my deep learning example and customer churn prediction using a support vector machine (SVM) as my machine learning example for this paper. I give an explanation of each approach's applicability for the issue and why the alternative does not.

Machine Learning Example: Customer Churn Prediction

Real-World Application

When a consumer stops doing business with a firm, such as when they cancel a phone plan in order to transfer carriers, this is known as customer churn. That is significant for telecom companies like Verizon, AT&T, and T-Mobile as their operations depend on monthly subscriptions. According to McKinsey, losing clients is also costly. They stated, “replacing the value of one lost customer can require the acquisition of three new customers” (O’Brien & Downie, n.d.). Omari et al. (2025) explains that, “accurate prediction models enable telecom companies to implement proactive strategies that reduce churn rates.” In the lesson's example, an SVM uses features like contract terms and monthly costs to learn from past information. Using a telecom dataset that “contains 7043 customer records with 21 features” (Section 3), Omari et al. investigated this and discovered that “SVM achieved the best results across all metrics” (Section

4). The SVM really “demonstrated highest accuracy (97%), followed by Logistic Regression (89%), K-Nearest Neighbors (89%), and Naïve Bayes (88%)" (Section 5).”

Why Machine Learning Is Suitable

The type of data is the main factor that makes standard machine learning effective in this situation. Because churn data is organized, it essentially sits in a spreadsheet where each row represents a client and each column represents something the business already keeps track of, such as contract type, monthly payment, and customer lifetime. These characteristics already have significance, and the company's executives are aware of which ones are important. “Contract type, monthly charges, and churn value are crucial for the prediction of customer churn”, according to Omari et al. (2025). It does not take a neural network to figure out that a client with a short term and a large bill would be considering leaving. Explainability is the other factor. The business needs to know why a client is at danger so it can take appropriate action, not simply a forecast. According to Omari et al., “by making predictive models interpretable, our aim is to provide telecom companies with actionable insights to improve customer retention strategies”. Additionally, simpler models may be retrained more quickly and affordably each month as new data becomes available.

Why Deep Learning Is Not Suitable

For this issue, deep learning would be unnecessary. Although churn data is not raw in that sense, its primary strength is the ability to learn features from chaotic, raw data, such as photos or sounds. There is nothing hidden for a deep network to extract because the helpful properties are already there in the columns. This is supported by research as Grinsztajn et al. (2022) found that “tree-based models remain state-of-the-art on medium-sized data (~10K

samples) even without accounting for their superior speed" after evaluating deep learning versus conventional models on "a typical collection of 45 datasets". With around 7,000 entries, the telecom dataset mentioned above is exactly in that territory. Additionally, deep learning has serious drawbacks for a company. "Deep approaches like ChurnNet" are mentioned in Section 1 of Omari et al. (2025), who also noted that "although ChurnNet achieves high performance, it requires significant computational resources and remains a 'black-box' model, limiting interpretability". The retention team has nothing to work with if a model predicts a customer's exit but is unable to provide an explanation on why.

Deep Learning Example: Facial Recognition

Real-World Application

I chose image recognition for deep learning, and Apple's Face ID is the real-world application I am most familiar with. I use it to unlock my phone hundreds of times a day, sometimes just after practice while wearing a hat, and it nearly always works. The TrueDepth camera operates "by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face" according to Apple (2026). Next, according to Apple's Advanced Technologies section, "a portion of your device's neural engine… transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data". The precision there is incredible. According to Apple, the system employs "advanced anti-spoofing neural networks" to prevent masks, and the likelihood of a stranger unlocking your phone is "less than 1 in 1,000,000 with a single enrolled appearance," (Apple, 2026). This is consistent with the lesson's example of a CNN identifying faces from unprocessed data without the use of traits like nose or eye shapes.

Why Deep Learning Is Suitable

Deep learning was designed with faces in mind. The input is raw picture and depth data rather than an organized table, and the patterns that distinguish one face from another are very complicated. This is handled by a CNN through layered learning. “Earlier layers focus on simple features, such as colors and edges," according to IBM (2021). The CNN begins to identify bigger parts or forms of the item as the visual input moves through its layers, eventually identifying the target object. The network does not need to know what a jawline looks like. Additionally, faces are constantly changing. According to Apple (2026), “Face ID automatically adapts to changes in your appearance, such as wearing cosmetic makeup or growing facial hair” and that it “is designed to work with hats, scarves, glasses, contact lenses, and many sunglasses”. Because a deep network learnt generic patterns rather than a set checklist, it can manage it. Since CNNs “can be computationally demanding,” the typical trade-off is processing power (IBM, 2021). Nevertheless, Apple incorporates a neural engine inside the device to manage this.

Why Traditional Machine Learning Is Not Suitable

Typical machine learning would suffer since it relies on human feature selection, which is nearly impossible when dealing with faces. The width of the nose and the distance between the eyes might be manually measured and sent into an SVM, but these measurements fluctuate with each grin, head tilt, and change in lighting. A checklist created by humans would fall apart as soon as someone moved their head and would overlook the majority of the minute characteristics that make a face distinct. CNNs were able to fix this precise issue. “Prior to CNNs, manual, time-consuming feature extraction methods were used to identify objects in images” according to

IBM (2021). Accuracy is crucial because Face ID is a security function as well. A hand-made checklist is unlikely to achieve Apple's one-in-a-million success rate.

Conclusion

I was able to better understand the difference between deep learning and machine learning after working through these two situations. For customer churn, a simpler model like an SVM is a better match since it utilizes structured data with attributes that customers are already familiar with, and the organization wants to know why a client could quit. The reverse is true with facial recognition. A CNN is the practicable option since the data is unprocessed, faces are constantly changing, and the key characteristics are too subtle to describe by hand. Returning to the lesson's analogy, churn prediction functions well with a checklist, but a student who has truly learned to read is required to recognize a face. Just knowing that deep learning is more sophisticated does not always make it superior. When the issue truly requires it, it is preferable.

References

Apple. (2026, September 18). *About Face ID advanced technology*. Apple Support. https://support.apple.com/en-us/102381

Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). *Why do tree-based models still outperform deep learning on tabular data?* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2207.08815

IBM. (2021, October 6). *What are convolutional neural networks?* https://www.ibm.com/think/topics/convolutional-neural-networks

O'Brien, K., & Downie, A. (n.d.). *What is customer churn?* IBM. Retrieved October 13, 2026, from https://www.ibm.com/think/topics/customer-churn

Omari, A., Al-Omari, O., Al-Omari, T., & Fati, S. M. (2025). A predictive analytics approach to improve telecom's customer retention. *Frontiers in Artificial Intelligence*, *8*, Article 1600357. https://doi.org/10.3389/frai.2025.1600357

Claude Opus 5.5 by Anthropic was used to help me conduct research, provide ideas of thoughts, and identify potential areas of importance that I should cover.